ERTS Lab
Department of Computer Science and Engineering
Indian Institute of Technology Bombay, Powai
Mumbai - 400 076.

Certificate of Participation

This is to certify that **Harshita Manish Puranik**, a student of **K. J. Somaiya College of Engineering, Maharashtra** has participated in the **e-Yantra Robotics Competition (eYRC 2025-26)**.

He/She is a member of the team having the following team members,

1. Kartavya Nilesh Kamble
2. Harshita Manish Puranik

This team has participated in **CropDrop Bot** theme.

Prof. Shivaram Kalyanakrishnan
Principal Investigator, e-Yantra
Associate Professor
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Indian Institute of Technology Bombay

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e-Yantra is a project sponsored by MHRD, Government of India, under the National Mission on Education through ICT (NMEICT).

Theme: *CropDrop Bot*

Theme Description:

The objective of this project is to design and implement an autonomous line-following and object-delivery robot, CropDrop Bot, that simulates agricultural logistics.

The robot begins its operation at a designated supply depot and autonomously navigates a predefined white-line track that represents farm pathways, including curves, zigzag sections, and discontinuous (dotted) segments. Along this route, the robot must detect and collect color-coded crates, each corresponding to a specific priority of the Delivery.Upon successful collection, the robot transitions to a black-line track that simulates village transportation lanes leading to Drop Zones. The robot must correctly identify the designated delivery zone for each crate

Participants are expected to design robust control algorithms with PID/RL to ensure reliable line tracking across varying path geometries, accurate color-based crate identification, precise pickup and drop actuation using electro-Magnet, and efficient task sequencing.

Theme Learning: *PID Control ,Reinforcement Learning, Control, Python, Embedded-C, Build-a-Bot.*

Certificate Level: The team has been awarded **Level 3** certificate in the e-Yantra Robotics Competition (eYRC).

> ***Level 3 indicates:*** *The team showed a fair understanding of the problem statement by implementing a partial solution of the problem statement.*

Certificate Level Matrix

Certificate Level	Certificate Description
Level 1	**Certificate of Merit**
Level 2	Certificate of Completion
Level 3	**Certificate of Participation**
Level 4	Letter of Participation

* Level 1 > Level 2 > Level 3 > Level 4 (Level 1 is highest achievement)